

Surrogate Modeling for Solving OPF: A Review

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Abstract: The optimal power flow (OPF) problem, characterized by its inherent complexity and strict constraints, has traditionally been approached using analytical techniques. OPF enhances power system sustainability by minimizing operational costs, reducing emissions, and facilitating the integration of renewable energy sources through optimized resource allocation and environmentally aligned constraints. However, the evolving nature of power grids, including the integration of distributed generation (DG), increasing uncertainties, changes in topology, and load variability, demands more frequent OPF solutions from grid operators. While conventional methods remain effective, their efficiency and accuracy degrade as computational demands increase. To address these limitations, there is growing interest in the use of data-driven surrogate models. This paper presents a critical review of such models, discussing their limitations and the solutions proposed in the literature. It introduces both Analytical Surrogate Models (ASMs) and learned surrogate models (LSMs) for OPF, providing a thorough analysis of how they can be applied to solve both DC and AC OPF problems. The review also evaluates the development of LSMs for OPF, from initial implementations addressing specific aspects of the problem to more advanced approaches capable of handling topology changes and contingencies. End-to-end and hybrid LSMs are compared based on their computational efficiency, generalization capabilities, and accuracy, and detailed insights are provided. This study includes an empirical comparison of two ASMs and LSMs applied to the IEEE standard six-bus system, demonstrating the key distinctions between these models for small-scale grids and discussing the scalability of LSMs for more complex systems. This comprehensive review aims to serve as a critical resource for OPF researchers and academics, facilitating progress in energy efficiency and providing guidance on the future direction of OPF solution methodologies.

Keywords: optimal power flow; surrogate model; power systems; Kriging; machine learning; neural network



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1. Introduction

Efficient distribution of electric energy is one of the critical issues when it comes to operating electric power grids. The grid operators have been facing this issue as part of their everyday decision-making duties. The way that this problem could be handled is to solve an optimization problem which provides the optimal current flow in the grid called optimal power flow (OPF). OPF is an optimization problem which is solved to minimize the cost of electricity generation subject to a series of physical and engineering constraints [1]. The physical constraints pertain to the limitations on generation units and transmission lines, while the engineering constraint involves maintaining power balance. AC OPF is the problem which covers all of the existing constraints and is solved to capture the detailed status of the power system. Generation scheduling, line congestion analysis, contingency analysis and even demand response studies are investigated by AC OPF solutions. OPF is a foundational tool in advancing sustainability within power systems, as it optimizes the dispatch and distribution of electrical power in ways that

minimize operational costs, reduce losses, and support the integration of renewable energy sources. By solving for the optimal power distribution under given system constraints, OPF facilitates reduced reliance on conventional, carbon-intensive generation methods, thereby lowering greenhouse gas emissions. Additionally, OPF enables systems to prioritize cleaner energy sources such as wind and solar energy by incorporating constraints and objectives related to environmental impact and resource efficiency. This optimization further assists in aligning power system operations with sustainability targets, ensuring that energy demands are met while upholding efficiency and resilience standards critical for sustainable power grids. Through this approach, OPF becomes instrumental in balancing economic objectives with ecological considerations, facilitating a transition toward low-carbon, reliable energy systems. The AC OPF, being a nonconvex problem, is complex due to a high number of equality and inequality constraints in large-scale grids, requiring nonlinear programming techniques to find the final solution. This problem is solved repeatedly every 5–15 min and it requires good availability of computational resources [2]. To tackle this concern, DC OPF, a simplified convex problem, is employed instead of AC OPF, in which a series of existing constraints are bypassed to reduce the complexity of the problem. Therefore, DC OPF is easier for grid operators to solve and also it requires less computational resources and time. Reliability assessment and market pricing are widely studied by DC OPF solutions. Under the market conditions, the uncertainty in the generation and demand response creates more complexity with regard to solving OPF [3,4]. Many traditional solvers have been presented to solve DC/AC OPF in recent decades, including analytical and heuristic methods. Among them, Newton–Raphson and Gauss–Sidel analytical approaches and Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) heuristic approaches have gained more attention [5]. Analytical approaches utilize gradient-based methods to iteratively find the final solution, whereas heuristic solvers draw inspiration from nature and human behaviors. In addition, there are numerous works that propose using linearized models to solve AC OPF using convex relaxation or approximation approaches [6–8]. It is possible to solve the linearized OPF using various linear programming methods without losing accuracy. Due to increasing uncertainty from integrating more Distributed Generation (DG), topology changes, and flexible loads, grid operators are required to solve OPFs more frequently. The well-known traditional methods can handle the OPF solutions very well, but if the frequency of running OPF is increased, these approaches need more time and cannot generate the accurate solutions in timely manner. Besides, the traditional methods are applied repeatedly to solve OPF in every single situation in the system. Therefore, there is a need to investigate more efficient ways to handle OPF in a short amount of time without ignoring accuracy.

Surrogate models provide a new opportunity to simplify OPF problems with high accuracy without carrying out iterative steps. A surrogate or metamodel is a simplified model which emulates an engineering or scientific problem with a high accuracy [9,10]. As computers become faster and more powerful, the surrogate is easier and cheaper to run compared to simulations and addresses the hardware limitations of existent computers. Thus, such models would be a proper alternative in many high-dimensional real-world problems like OPF. Optimal power flow is a complicated nonlinear problem with lots of unknown variables, considering partial derivatives and complex numbers calculations, that should be solved in short intervals for accurate and optimal power system operation. However, for large-scale power systems, solving OPF takes a long time, which is not reasonable in the presence of fast load profile changes. Thus, surrogate modeling can solve this problem by speeding up the process and providing almost real-time solutions for OPF problems. Learned surrogate models can also provide OPF solutions for contingencies and topology variations based on the trained model without additional computations. As discussed in the next sections, learned surrogate models (LSMs) can provide OPF solutions for contingencies and topology variations based on the trained model without additional computations. Table 1 shows the main differences between traditional OPF solvers and data-driven surrogate-based solvers.

Table 1. Comparison between traditional OPF solvers and data-driven surrogate models.

Feature	Traditional OPF Solvers	Data-Driven Surrogate Models
Computational Speed	Relatively slow, especially for large networks with many constraints	Fast inference after training, suitable for real-time applications
Scalability	Limited by solver complexity, may struggle with large-scale networks	Scalable once the model is trained, no need to solve optimization problems repeatedly
Accuracy	High, depending on the complexity and resolution of the model	Can approximate well, but accuracy depends on training data quality and model capacity
Training Requirement	No training required; directly based on system model	Requires a large amount of historical data for training
Adaptability to Changes	Rigid; requires reformulation for new network configurations or constraints	Flexible; can be retrained with new data to adapt to changing conditions
Handling Uncertainty	Can handle uncertainty explicitly via stochastic or robust optimization techniques	Handles uncertainty implicitly if the training data cover enough variability

In this review paper, surrogate models have been reviewed for both DC and AC OPF problems in power systems. Analytical (ASMs) and learned surrogate models (LSMs) are introduced for the first time to provide a better comparison and categorization of the presented methods. To the best of the authors' knowledge, this is the first paper which presents a comprehensive analysis of surrogate models for OPF problems. In [11], only traditional and heuristic methods were investigated. Machine learning (ML)-based methods in OPF is presented in [12], where they are not introduced from the surrogate model point of view. A brief introduction of ML methods for OPF was also presented in [13], which does not provide deep considerations for the analysis of different methods and the surrogate concept is also absent. A tutorial is presented in [14], which investigates different ML-based approaches to the OPF problem and provides some simulation-based results to demonstrate the existing differences. However, it only reviews 45 papers and does not involve the surrogate concept. The applications of machine learning in AC OPF are reviewed in [15], in which DC OPF is not addressed, the surrogate concept is absent, and the main categories are not based on the inherent challenges of proposed methods for OPF. Thus, this paper provides a comprehensive review of OPF surrogates based on new insights reviewing the most recent published papers. The findings of this review could be employed to investigate new avenues in OPF surrogate modeling for both the academic and industrial sectors. The main contributions of the paper are as follows:

- This paper analyzes the state-of-the-art methods for surrogate models of both ASMs and LSMs in the OPF problem for the first time.
- Simulation results are presented to visualize some of the key features of both surrogate models for the OPF system. The IEEE six-bus system is considered as a standard case study, and two analytical and two learned surrogates are considered for comparison to show the differences for building OPF surrogates.
- The review process aims to demonstrate how the featured works tackle various concerns and challenges in OPF surrogate modeling that have not been previously addressed.
- This paper also provides the key aspects, recommendations, and scope for future work to help beginners and researchers gain exposure to the OPF surrogate model and its improvements.

The remainder of the paper is organized as follows: Section 2 discusses the OPF formulations. Surrogate modeling is examined in Section 3. Section 4 introduces ASMs, while LSMs are presented in Section 5. Simulation results comparing ASMs and LSMs are provided in Section 6. Section 7 covers challenges and future directions. Finally, the conclusion is presented in the last section of the paper.

2. OPF Formulation

OPF is an optimization problem designed to minimize the costs of electricity generation while accounting for physical and engineering constraints. These constraints include generation limits, transmission line capacities, load demand, and operational policies. They are typically represented as nonlinear equations or inequalities, requiring nonlinear programming techniques to solve the OPF accurately. However, simplifying these nonlinear terms results in a linear optimization problem, which can be solved more quickly using linear programming methods. This approach, though faster, introduces some errors due to the omission of critical constraints. The general DC/AC OPF formulations are as follows [16]:

$$\min \sum_{i \in \mathcal{G}} c_i P_{G_i} \quad (1)$$

DC OPF constraints:

$$P_{G_i} - P_{D_i} = \sum_{(i,j) \in \mathcal{L}} B_{ij}(\theta_i - \theta_j), \quad \forall i \in \mathcal{N} \quad (2)$$

$$P_{G_i}^{\min} \leq P_{G_i} \leq P_{G_i}^{\max}, \quad \forall i \in \mathcal{G} \quad (3)$$

$$-P_{ij}^{\max} \leq B_{ij}(\theta_i - \theta_j) \leq P_{ij}^{\max}, \quad \forall (i,j) \in \mathcal{L} \quad (4)$$

AC OPF constraints:

$$P_{G_i} - P_{D_i} = \sum_{j \in \mathcal{N}} |V_i||V_j| \times (G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j)), \quad \forall i \in \mathcal{N} \quad (5)$$

$$Q_{G_i} - Q_{D_i} = \sum_{j \in \mathcal{N}} |V_i||V_j| \times (G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j)), \quad \forall i \in \mathcal{N} \quad (6)$$

$$V_i^{\min} \leq |V_i| \leq V_i^{\max}, \quad \forall i \in \mathcal{N} \quad (7)$$

$$P_{G_i}^{\min} \leq P_{G_i} \leq P_{G_i}^{\max}, \quad \forall i \in \mathcal{G} \quad (8)$$

$$Q_{G_i}^{\min} \leq Q_{G_i} \leq Q_{G_i}^{\max}, \quad \forall i \in \mathcal{G} \quad (9)$$

$$|S_{ij}| \leq S_{ij}^{\max}, \quad \forall (i,j) \in \mathcal{L} \quad (10)$$

and the parameters for both DC and AC OPF are defined in Table 2. The surrogate models are proposed to consider the OPF problem within the existent hard constraints. ASMs use interpolation techniques to find the OPF solution in the range of available data in the training process, while LSMs utilize machine learning approaches to predict OPF solutions for unseen data. In the next few sections, a detailed discussion of how both methods are implemented is provided.

Table 2. Summary of DC and AC OPF Parameters.

Symbol	Description	Units
c_i	Cost coefficient of real power generation at bus i	\$/MW
P_{G_i}	Real power generation at bus i	MW
P_{D_i}	Real power demand at bus i	MW
Q_{G_i}	Reactive power generation at bus i	MVar
Q_{D_i}	Reactive power demand at bus i	MVar
B_{ij}	Imaginary component of the admittance matrix, representing the susceptance of the line connecting buses i and j (for $i \neq j$)	S (Siemens)

Table 2. Cont.

Symbol	Description	Units
G_{ij}	Real component of the admittance matrix, representing the conductance of the line between buses i and j (for $i \neq j$)	S (Siemens)
θ_i	Phase angle of the voltage at bus i	rad
$ V_i $	Magnitude of the voltage at bus i	kV
$p_{ij}^{\max}$	Maximum real power flow on line (i, j)	MW
$s_{ij}^{\max}$	Maximum apparent power flow on line (i, j)	MVA
$\mathcal{G}$	Set of generators	-
$\mathcal{N}$	Set of buses or nodes	-
$\mathcal{L}$	Set of transmission lines	-

3. Surrogate Modeling

Surrogate model applications are now evident in many fields, such as chemical engineering, aerodynamic design, building design, water management systems and other areas [17–21]. For instance, when modeling the reactions in an internal combustion engine, it takes a huge amount of time to consider every single reaction, so a surrogate model could be a good alternative to simulate the overall process in much less time. There are different types of surrogate models that have been proposed in the past few decades. The modeling techniques can be broadly categorized into two main groups: ASMs and LSMs. These groups are briefly defined in the following subsections and shown in Figure 1. In the ASM branch, the model construction relies on well-established mathematical formulations to approximate complex functions. This category includes interpolation techniques, such as radial basis functions and Kriging, which are widely used to approximate values between known data points based on spatial relationships. Additionally, regression methods, including linear regression and support vector regression, aim to model the relationship between input variables and outputs by minimizing the error between predicted and actual values. Mixture methods are also included, which likely combine different techniques to enhance surrogate model accuracy. The LSM category encompasses data-driven approaches for surrogate model development, leveraging the power of machine learning. Supervised learning utilizes labeled datasets to train models that can predict outcomes for new, unseen data, while unsupervised learning aims to uncover hidden patterns within unlabeled data. Self-supervised learning operates in an intermediate space, generating labels from the input data itself, often for tasks like representation learning. Finally, reinforcement learning involves learning optimal actions through interaction with an environment, using a reward-based system to iteratively improve performance.

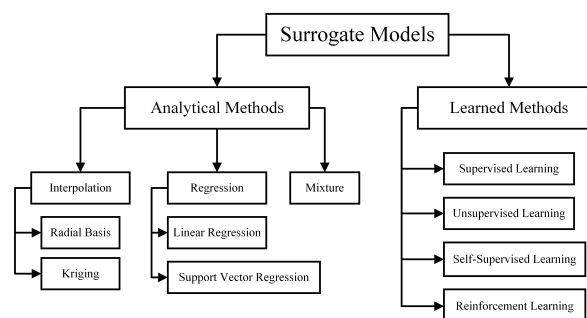


Figure 1. Surrogate modeling methods: The analytical branch explores techniques such as interpolation (e.g., radial basis and Kriging) and regression (e.g., linear and support vector regression), while the Learned Methods branch delves into advanced paradigms, including supervised, unsupervised, self-supervised, and reinforcement learning strategies.

3.1. Analytical Surrogate Models

ASM is widely utilized for many complex and time-consuming engineering/industrial problems. The structure of a typical ASM is depicted in Figure 2. It comprises several stages, as follows:

- **Design parameters:** The sampling method that represents the selected problem along with the number of samples.
- **Simulations/experiments:** Computer simulations or real tests are performed to collect the required dataset.
- **Dataset collection:** Based on the simulations/experiments, an efficient number of samples that shows the mapping between input and output of the problem are collected. Some pre-processing techniques are applied for data cleaning purposes.
- **Model validation:** The effectiveness of the surrogate model is analyzed and the termination rule is indicated to extract the proper model.

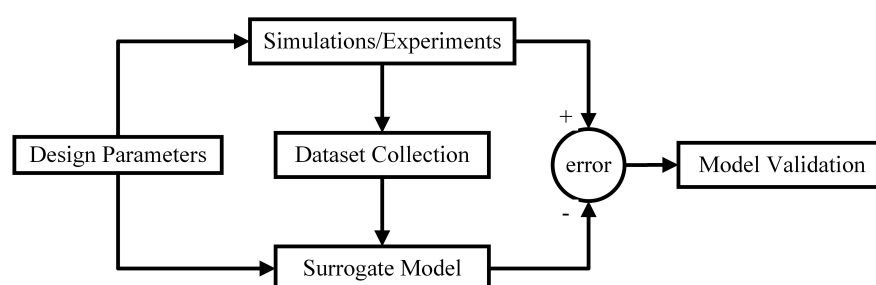


Figure 2. A typical structure of ASM. The diagram shows an iterative process in which design parameters drive simulations, resulting in dataset collection for surrogate modeling. The model is validated by assessing error in a feedback loop.

There are three main groups for this type of metamodel:

- **Interpolation methods:** The mapping errors between input and output data are minimized in a range of training data, which means that the predictions are restricted by the range of training data. Radial basis function and Kriging are categorized in this category.
- **Regression methods:** The mapping errors between input and output data are minimized in the range and outside of training data. Linear regression and support vector regression are categorized as regression-based methods. An optimization problem is solved to capture the surrogate model based on a vector of weights as well as a basis function.
- **Mixture:** A hybrid method that can be applied in very complicated problems would be more complex with higher accuracy.

3.2. Learned Surrogate Models

In recent years, ML is increasingly applied in various fields and plays an important role in real-world applications, including object recognition, stock market prediction, weather prediction, image processing, and even emerging chatbot applications that use new generative concepts [22–26]. So, this new concept would be a promising option for building precise surrogate models. Datasets for complex systems can be generated through simulations, experimental tests with predefined inputs and outputs, or by utilizing large historical data from precise sensors in industrial sectors. The ML models are then trained to accurately map these inputs to outputs, offering a faster alternative to traditional simulations or tests. These models can simplify real-world problems with high accuracy. Furthermore, models can be developed without ground truth by using a well-defined loss function that encompasses the system's physical and engineering constraints. In this paper, these types of metamodels are called LSMs. Four main groups exist, as follows:

- **Supervised learning:** An LSM is trained based on the labeled data that represent the relevant input/output pairs.
- **Unsupervised learning:** An LSM is trained based on the unlabeled data from which the correlated groups and clusters are extracted.
- **Self-supervised learning:** An LSM is trained based on the unlabeled data; the goal is to find the best labels for the input data.
- **Reinforcement learning:** An LSM is trained based on the interaction of an agent with the environment to maximize specific predefined rewards based on received feedback.

Table 3 summarizes the main differences between ASMs and LSMs based on main features. LSMs are able to make highly accurate predictions about unseen data and can handle a large amount of data with high generalization. Therefore, the majority of presented methods have utilized LSMs for the OPF problem, which is the main focus of this review paper.

Table 3. Analytical surrogate models versus learned surrogate models.

Features	ASM	LSM
Complexity	Low	High
Data Requirement	Low	High
Interpretability	High	Low
Generalization	Low	High
Scalability	Low	High
Robustness	Low	High

4. Analytical Surrogate Models for OPF

The Kriging method, as the most effective ASM technique in the presence of uncertainties, is proposed to solve the OPF as presented in [27–34]. The Kriging method uses prior data covariance as a linear interpolation method and can provide the best unbiased predictions [35]. At first, the labeled OPF solutions should be generated based on simulations or prepared via past historical data. Then, Kriging will be utilized to interpolate the OPF solutions. In this way, each entered datum will improve the accuracy of the surrogate model. Uncertainty about the renewable energy sources could be handled using the Kriging method by providing a range of OPF solutions. Kriging models the function $f(\xi)$ mapping inputs to outputs as follows:

$$f(\xi) = \mu(\xi) + Z(\xi) \quad (11)$$

where $\mu(\xi)$ is the deterministic trend function (often assumed to be constant or linear) and $Z(\xi)$ is a zero-mean Gaussian process representing deviations from the trend.

The covariance between two points is defined as follows:

$$\text{Cov}[Z(\xi_i), Z(\xi_j)] = \sigma^2 R(\xi_i, \xi_j) \quad (12)$$

A common choice for the correlation function is the Gaussian correlation:

$$R(\xi_i, \xi_j) = \exp\left(-\sum_{k=1}^d \theta_k |\xi_{i,k} - \xi_{j,k}|^{p_k}\right) \quad (13)$$

where θ_k are non-negative parameters controlling the correlation decay rate, p_k is the exponent parameter (usually $p_k = 2$ for Gaussian correlation), and d is the number of input dimensions.

Given n training data points $\{(\xi_i, y_i)\}_{i=1}^n$, the Kriging predictor at a new point ξ^* is

$$\hat{y}(\xi^*) = \hat{\mu} + \mathbf{r}^\top(\xi^*) \mathbf{R}^{-1}(\mathbf{y} - \hat{\mu}\mathbf{1}) \quad (14)$$

where $\hat{\mu}$ is the estimated trend (mean) of the process, $\mathbf{r}(\boldsymbol{\zeta}^*)$ is the correlation vector between $\boldsymbol{\zeta}^*$ and training inputs, $\mathbf{R}$ is the correlation matrix of training inputs, $\mathbf{y}$ is the vector of observed outputs, and $\mathbf{1}$ is the vector.

The predicted variance is

$$s^2(\boldsymbol{\zeta}^*) = \sigma^2 \left[1 - \mathbf{r}^\top(\boldsymbol{\zeta}^*) \mathbf{R}^{-1} \mathbf{r}(\boldsymbol{\zeta}^*) + \frac{(1 - \mathbf{1}^\top \mathbf{R}^{-1} \mathbf{r}(\boldsymbol{\zeta}^*))^2}{\mathbf{1}^\top \mathbf{R}^{-1} \mathbf{1}} \right] \quad (15)$$

So, based on the above-mentioned formulations, the wind power or solar generation can be considered as the data with uncertainty.

ASMs Based on the Kriging Method for Solving OPF

Due to its strong and powerful features, the Kriging method is the only ASM method which has been investigated in past works for solving the OPF problem. AC OPF ASM for power systems in the presence of wind farms are investigated in [27]. In [28], the Kriging method is introduced to assist heuristic methods like GA and PSO to enhance the speed and accuracy. In [29], a Kriging-aided surrogate is presented for chance-constrained (CC) DC OPF considering *affine policy*. The uncertainty of DGs is considered in the CC OPF. In [30], a surrogate model is employed to handle AC OPF in a distribution network considering DG, battery storages, and interruptible loads. A PSO optimization-based methodology is used to handle the optimization part of the Kriging-based surrogate model. A similar idea is implemented in [31] considering a diversity of controllable resources, including energy storage devices, DGs, voltage regulators, and switchable capacitor banks. The paper shows that the proposed method provides a better solution than GA and PSO within the predefined expensive evaluation time limit. In [32], the possibility of applying Kriging-based surrogates for large-scale power systems is investigated by analyzing an IEEE 300-bus test system. The CC AC OPF is the objective of the paper. In [33], Gaussian process or Kriging method is utilized to build a surrogate model. The sparse and hybrid Kriging method is applied to present a fast surrogate model for solving AC OPF in [34].

In general, ASMs encounter several challenges, particularly regarding the appropriate selection of modeling methodologies and the application of precise sampling techniques. A significant concern is the curse of dimensionality, which results in an exponential growth in the number of required training points as the problem's dimensionality increases, thereby complicating the development of an accurate surrogate model. This phenomenon adversely impacts the performance of most ASMs, leading to substantial degradation in high-dimensional settings. Additionally, the Kriging method has been evaluated on a 300-bus test system, though only under worst-case conditions [36].

5. Learned Surrogate Models for OPF

The limitations of ASMs have motivated researchers to investigate more appropriate methods. The LSM could provide more accurate surrogate models by incorporating learning-based concepts. Numerous studies have verified this claim. Two general approaches have been studied for LSMs, as presented in Figure 3. The first one is the end-to-end approach, which is a direct mapping to generate the OPF solutions. In this approach, active/reactive demands are the inputs and OPF solutions, offering active/reactive generations, and voltages are the outputs. As shown in Figure 3, in an end-to-end method, the final OPF solutions are predicted via the trained model without any external calculations. In other words, the LSMs emulate the OPF problem considering all equality and inequality constraints. The supervised learning approach, favored for its ability to utilize input/output data from simulations or historical grid data, faces challenges with regard to satisfying the OPF problem's constraints due to the opaque nature of NNs. This method requires a substantial labeled dataset to cover all OPF scenarios, which may not always be available due to insufficient simulations or unrecognized grid features. Alternatively, a hybrid method employs LSMs for a portion of the OPF problem, with remaining variables

determined through predictions and a specialized reconstruction scheme. This approach incorporates physical OPF constraints directly into the reconstruction phase, utilizing traditional solvers to finalize the solution. Although this might extend the solution time compared to end-to-end methods, hybrid methods offer more generalized solutions by integrating problem constraints into the training, either by modifying the NN architecture or adjusting the loss function, enhancing overall adaptability to the OPF challenges. In the upcoming sections, different methods that are generally categorized into two groups are considered for a comprehensive review. Some basic directions are designed to create a logical trend demonstrating how the main concerns have been responded to by the presented LSMs for OPF. In this way, the main challenges and questionable parts are discussed based on a step-by-step approach.

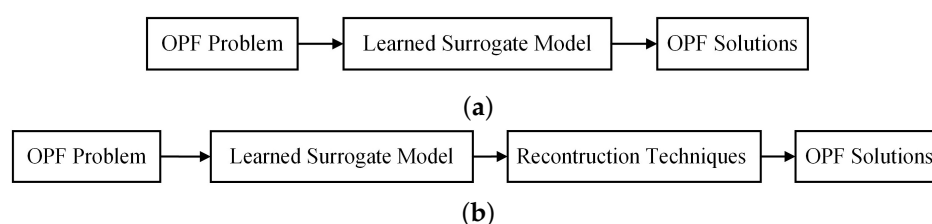


Figure 3. LSM approaches for solving OPF: (a) end-to-end (b) hybrid.

5.1. Typical LSMs for OPF

OPF traditional solvers like the Newton–Raphson solver implement partial differentiation to obtain the optimal parameters, which is the main reason that the process of iterative solvers is so time-consuming. In [37], a novel iterative method is introduced that uses only one hidden layer to generate the optimal solutions for the AC OPF problem as a new surrogate model. In fact, the NN acts like an iterative solver and the inputs are updated based on the output feedback. A labeled dataset is required that represents the optimal solutions, and then the NN learns how to map the inputs with the desired outputs by minimizing the prediction error, which is same as the error that exists in traditional iterative methods. In another work, the quasi-Newton method is simplified by implementing an NN in [38] that removes the need to calculate Jacobian matrices. Thus, using an NN as a successor could elevate the calculation time exponentially. A one-layer convolutional NN (CNN) is implemented as an end-to-end approach to estimate DC OPF for IEEE 14, 118, and 300-bus test systems, as presented in [39]. The same approach is proposed in [40], in which CNN presented better results than a simple NN, so the extracted surrogate model will represent the changes in the grid structure. A multi-target random forests model is proposed in [41] to predict the multiple unknown parameters of the AC OPF without any information about the grid topology. In [42], an NN-based classifier is used to indicate the feasible DC OPF solution for different input scenarios. Then, if the considered scenario is feasible, another NN is trained to find OPF solutions without any constraint violations. These methods provide a straightforward end-to-end solution for OPF; however, they do not necessarily guarantee zero violation of the hard constraints that exist. The surrogate model training error increases linearly as the grid scale becomes larger.

5.2. Increasing the Generalization

As discussed, generalization is one of the main challenges that arises when building efficient surrogate models for OPF problems with a high number of hard constraints. Plenty of researchers have attempted to increase the generalization of the proposed LSMs in both DC and AC OPF problems, which are categorized in multiple groups. Table 4 shows a summary of the methods which are categorized in this group.

5.2.1. Finding Active Constraints

A group of presented solutions has focused on active sets of constraints in DC OPF problems [43–46]. Active sets indicate the critical constraints in OPF problems that have a

direct impact on the optimal solution. In [43], the NN structure is constrained by checking the Karush–Kuhn–Tucker (KKT) condition and also, to ensure that a fully convex problem was designed, ReLU and Sigmoid activation functions were not utilized in the training process. ReLU and Sigmoid are nonlinear in nature. The number of active set constraints is exponentially increased when the size of the system is increased. In [44], *ensemble policy* is used to find the relevant optimal active sets. However, this approach is applicable only for a low number of active sets. To address this issue, a classification algorithm is introduced to find the relevant active set of constraints in [45]. An enhanced preventive learning approach is presented in [46] where non-critical inequality constraints are removed, the equality constraints are ignored, and a calibrated ground truth method is considered in which a set of harder constraints is provided to guarantee the optimization feasibility. In other words, the minimum and maximum bounds of decision variables are changed to increase the solution feasibility without violating OPF constraints.

Table 4. Summary of Proposed Methods for Increasing the Generalization.

Method	Approach	Key Aspects
Finding active constraints [47–49]	Considering only critical constraints in OPF problems	increased feasibility, computational cost is decreased, complexity of finding all critical constraints for large-scale grids
Scaling Factor [50–52]	Converting the inequality constraints to equality constraints	zero constraint violation, extra calculations needed to obtain the final OPF solution
Physics-Informed Methods [53–55]	Utilizing OPF physical constraints in training process	feasibility increased, complexity is increased due to upgraded loss functions
Split-wise Methods [47–49]	Handle a part of OPF to simplify the problem	less training time, extra calculations are needed for final OPF solutions
Miscellaneous Methods [56–59]	Different approaches are proposed	Higher feasibility, robustness against uncertainty

5.2.2. Implementing Scaling Factor

Inequality constraints are handled with an innovative approach in [50–52]. A scaling factor is used to reform inequality constraints to equality constraints and then the scaling factor is estimated by the trained model instead of the OPF optimal solutions. Linear scaling can be used to convert the inequality constraints to equality constraints. The general formula for the conversion is as follows [51]:

$$\begin{aligned} x_{min} &\leq x_{desired} \leq x_{max} \\ x_{desired} &= \alpha(x_{max} - x_{min}) + x_{min} \end{aligned} \quad (16)$$

where $x_{desired}$ can be the constrained variable in our problem. Thus, the aim of the neural network is to predict the α coefficients instead of $x_{desired}$. Using this technique guarantees that there will be zero constraint violations. The α coefficient is always between 0 and 1, so the sigmoid activation function would be the best choice for the output layer. In case of OPF problem inequalities for generation power, line congestion and bus voltages can be converted to equalities utilizing this technique. Thus, in these supervised methods, zero violation of inequality constraints is guaranteed. These supervised approaches are listed in hybrid methods since further calculation is needed to obtain the final solution.

5.2.3. Physics-Informed Approaches

The physical condition of the power system is defined as a portion of constraints in the OPF problem. In order to increase the generalization and reduce the violations, one can model these physical constraints in the LSM training process. Designing a new loss function or changing the layers in the NN could be categorized in this group. Physics-informed NNs (PINNs) are employed in [53,54] in which the loss function of the NN is modified

by adding constraint violations. Three NNs for estimating power generation, voltage and dual variables are trained. The KKT condition is considered in [53] to check the optimal solutions during the training process. In [54], the trained model predicts the generation values and violations on physical constraints simultaneously. The predicted voltages are modified by a predefined function to ensure constraint satisfaction. Reconstructing the NN architecture is implemented in [55] via a projection-aware method to obtain zero violation of constraints. The projection-aware layer is considered between the last hidden layer and output layer and enforces the output of the last hidden layer to be in the acceptable range. Back-propagation is utilized by considering a new scheme which increases the complexity. PINNs are supervised end-to-end methods that could decrease the violations as well as surrogate model training error.

5.2.4. Splitwise Approaches

In splitwise approaches, which are a part of hybrid methods, a part of the OPF problem is handled with LSMs to simplify the overall problem. In this way, the computational cost and constraint violations are decreased and feasibility is increased. Finding the AC OPF surrogate by utilizing multiple NNs is the main aim of the methods described in [47–49]. Three multilayer perceptrons are trained to find the current safe distance, voltage safe distance and total power loss in [47]. In fact, the proposed method in this paper estimates the violations of an AC OPF problem by three NNs, and then mixed-integer linear programming is used to find the optimal solutions of AC OPF without constraint violations. In [48], a similar idea as that in [47] is utilized to train two multilayer perceptrons for physical constraint violations prediction. Then, the AC OPF problem is solved in the absence of both equality and inequality constraints. A new idea, activation regions, is presented in this paper to filter some undesired input samples in the training process. The same idea is utilized in [49] to solve CC AC OPF considering the uncertainty of renewable energy sources. The methods presented in [47–49] mitigate the need to access grid topology in distribution networks. As grid parameters in distribution networks are not fully monitorable, these approaches have proposed efficient surrogates in less observed grids.

5.2.5. Miscellaneous Methods

There are other proposed methods that employ various schemes to address increasing generalization. In [56], the holomorphic function is substituted for the AC OPF equation, which is more likely to find existing OPF solutions. In [57], a post-processing section is added to enforce a supervised NN-based surrogate AC OPF solver's outputs into the desired ranges. In other words, if the outputs have violations, the variables are modified with acceptable values. In [58], a *model-informed generative adversarial network* is applied to handle the uncertainty in the AC OPF problem. Adversarial data were used to increase the robustness of the NN in [59].

5.3. Handling Large Datasets

Generating a big labeled dataset is time-consuming and is not applicable to every situation, especially when limited historical data are available. Therefore, investigating new methods that need a lower/bounded number of data is necessary. There are three types of solution that have been presented in the past, including reducing/bounding input/outputs of the trained model, implementing unsupervised or self-supervised methods, and implementing reinforcement learning that does not use any labeled data. Table 5 shows a summary of the methods which are categorized in this group.

Table 5. Summary of proposed methods for handling large datasets.

Method	Approach	Key Aspects
Reducing input/output size [60–65]	Eliminating the demand data that cause infeasibility	increase the feasibility, computational cost is decreased, infeasible load profiles should be evaluated as the data preprocessing and this increases the computational time and cost
Unsupervised and Self-supervised Methods [66–71]	Finding OPF solutions without predefined data, incorporating all OPF constraints in the NN loss function	no need to labeled synthesised or historical data, no guarantee of feasibility
Reinforcement learning methods [72–80]	Reinforced NN-based LSMs considering agents	feasibility increased, constraint violations decreased, training time increased

5.3.1. Reducing Inputs/Output Size

In terms of large-scale power grids, there are a large number of variables which must be considered during the training process. Load profiles differ with the input and generation rates and bus voltages as the outputs can be very large. This reduces the trained surrogate model, especially for large outputs, since the prediction accuracy will be lower.

Compact learning is used in [60] to handle the end-to-end surrogates. This method uses *principal component analysis* during the training process to narrow down the input/output size of the trained model. In [61], a novel teacher–student method is used to decrease the dependency on big data to find a surrogate model for DC OPFs with lower complexity. Feasible solutions involving a smaller amount of data are utilized in [62] with conditional generative adversarial networks to train a model to solve AC OPF without knowledge of the grid topology. Latin hypercube sampling is used instead of random sampling in [63] to consider only uncorrelated scenarios when generating data for load values. Then, the gamma distribution function is used along with a proper threshold to cancel the low probability constraints in the problem. Initial values of active/reactive generation and voltages along with the grid demand are considered as the inputs of the LSM in [64]. A multi-target decision tree is proposed in [81] to find the optimal warm-start condition for AC OPF. In other words, load profiles with higher feasibility are indicated by implementing the decision tree classifier. In [65], a framework for multi-task learning is proposed to speed up the AC OPF traditional solvers by predicting the solutions.

5.3.2. Unsupervised and Self-Supervised Methods

The lack of sufficient historical data and the time-consuming process of generating labeled data have motivated researchers to investigate unsupervised or self-supervised learning approaches. In [66], partial solutions are introduced for equality constraints and gradient-based solutions are corrected to satisfy inequality constraints. The primal–dual (PD) method is applied as a self-supervised learning method without any ground truth in [67]. The primal problem is the main optimization problem with all of the constraints, which in our case is AC OPF. The dual problem is another optimization problem based on the primal problem with a specific objective and constraints. The PD uses a penalty-based loss function based on the AC OPF constraint violations. Another self-supervised method is proposed in [68] that uses the same concept as [67]. Stochastic primal–dual updates are employed to learn the trained weights. A loss-guided NN is employed to learn AC OPF optimization without any ground truth and performed in [69]. All of the problem constraints plus the generation cost are converted to a loss function and the training process goal is to minimize the defined loss function. The main challenge in using loss-guided NNs is obtaining the gradients via the back-propagation method. An unsupervised method in [70] is proposed in which DC OPF is solved by considering the Lagrangian duality. This method is a hybrid method that uses the predicted parameters to find the optimal OPF solution. The method in [71] presents a self-supervised method to generate a surrogate

for a DC OPF problem without any ground truth, in which $N - 1$ contingency along with demand uncertainty is considered.

5.3.3. Reinforcement Learning Methods

Reinforcement learning (RL) methods are presented to mitigate the need for labeled data in supervised learning and to enhance unsupervised learning methods by implementing the agent concept [72–80]. Reinforcement learning can be utilized along with the surrogate model to provide more efficient outputs by interacting the agents with the surrogate model. In case of optimization problems like OPF, the agents can guide the surrogate model to minimize the constraint violation and enhance the feasibility and generalization.

Learning by demonstration is used in [72] as a modified deep reinforcement learning method to enhance the feasibility. A real-time surrogate model based on gradients of the AC OPF problem along with reinforcement learning is proposed in [73]. The critic part of reinforcement learning is being replaced with the augmented cost of the AC OPF problem, which indicates the effectiveness of the selected actions during the process by considering the physics of the problem. Reinforcement learning is introduced to enhance the *distributed alternating direction method of multipliers* by finding dynamic penalty parameter selection in [74]. In [75], a method called *deep deterministic gradient policy* is used to speed up the reinforcement-based surrogate model by modifying the critic network. Uncertainty of the loads in a power system is considered in [76]. In [77], reinforcement learning and traditional optimization algorithms are presented to indicate inactive constraints in an AC OPF problem, which increases the feasibility and generalization. A primal–dual approach along with reinforcement learning is introduced to increase the generalization of a reinforcement-based surrogate model in [78]. In [79], a surrogate model using deep reinforcement learning is proposed to solve AC OPF for the distribution system.

5.4. Decentralized LSMs for Large-Scale Power Grids

Decentralized OPF was investigated to solve OPF problems in large-scale power systems with a minimum volume of communication [82,83]. This approach would be a good choice to implement LSMs for OPF in such large, interconnected grids [84–86]. In [84], the power system is divided into some independent sections and a decentralized approach is introduced to solve DC OPF. This method uses an NN to accelerate the solving process, not to find optimal solutions, and this was the first paper to investigate a distributed solution to find a DC OPF solution. An NN is trained to find line coupling and voltages for multiple independent regions in power system in [85]. Then, a specific number of NNs are trained to solve AC OPF as an end-to-end scheme for each region. While this method decreases the training time for large-scale power systems, a set of multiple NNs must be trained, which increases the complexity, especially with regard to finding different separate regions. A decentralized approach based on multi-agent systems is developed in [86]. This method uses a centralized optimization method to generate the inputs for each agent and then decentralized learning is implemented to find the optimal solutions for the problem.

5.5. Grid Topology Changes and Contingencies

Two of the main challenges in the OPF optimization problem are the contingencies and topology changes. While many security-constrained solutions have been proposed to address contingencies in the past [87], there is no solution for traditional solvers to consider topology changes in the power grids. In this section, the LSMs that are presented to tackle topology changes and contingencies are investigated.

5.5.1. Graph Neural Network (GNN) Methods

The GNN is one of the most efficient choices for employing LSMs to solve OPF problems in variable configuration grids. A graph can represent the grid structure in a way that considers the possible configuration status. Thus, it provides prominent features for addressing the mentioned challenges [88–96].

Imitation learning, as a centralized method, is proposed in [88] to enhance the scalability and feasibility of the GNN surrogate model. However, this method is not applicable in large power systems. An unsupervised GNN-based method that does not need labeled data is presented in [89]. The penalty function based on the violations of OPF constraints is defined to guide the GNN training process. Another GNN-based method which uses locational marginal prices and voltages as estimated variables is employed in [90]. Also, topology variations and $N - 1$ contingency are investigated to validate the proposed methodology. Two recent papers utilized GNN-based models to consider topology changes in the grid to propose a more generalized surrogate model for unseen grids [91,92]. The GNN-based models offer a better opportunity to consider different grid topologies, but they do need a huge labeled dataset for prediction. The method presented in [95] used a guided drop-out method to consider $N - 1$ contingency in the training process. In this way, some of the neurons of hidden layers were controlled externally. In addition, GNN was employed with a modified loss function that covered the constraints of the problem. Graph CNN with long short-term memory is adopted in [96] to handle both topology changes in the grid and uncertainties of the renewable energy resources. In [97], a physics-guided GCNN method is applied to consider topology changes in the grid. A physics-guided graph convolution kernel, feature construction, and loss function formulation are designed simultaneously.

5.5.2. Miscellaneous Methods

Apart from the GNN, there are other different methods that consider topology changes in the OPF surrogate extraction. In [98], the proposed method combined different topology input data and activation functions to obtain a more generalized model. The training time for large inputs was very high. Discrete topology representation along with active/reactive demand were adopted as inputs of the LSM to present a grid-aware method in [99]. A security-constrained AC OPF is investigated in [100] to mitigate the contingency cases that have no effect on the optimal solution to the problem. The proposed method provides a simpler surrogate model in case of topology uncertainty. In [101], CNN is considered to learn AC OPF with the topology labels in input data. In [102], a decentralized approach is implemented to investigate a surrogate AC OPF in a distribution system with time-varying topology changes of the grid. The AC OPF was emulated in an initial 31-bus system and then expanded to a 59-bus system. Table 6 provides a brief summary and key aspects of LSM methods in OPF.

Table 6. Summary of different OPF LSMs.

Proposed Methods	Key Negative Aspects
Basic methods	Low generalization; low complexity
Methods for increasing the generalization	Large data requirement, low constraint violations, high complexity
Methods for handling large datasets	Need for reconstruction, lower generalization, RL challenges in cold-start condition
Methods for decentralized LSMs	Cybersecurity challenges, high complexity, large data requirement
Methods considering topology changes and contingencies	Large data requirement, long training time, need for prior knowledge of the grid

6. Comparison of Analytical and Learned OPF Surrogates via Simulation Results

In this section, a simulation-based comparison is provided by considering two LSMs (artificial NN (ANN) and CNN) and two ASMs (linear regression and support vector regression) to reveal some preliminary differences between learned and analytical methods. In fact, they are selected as toy models to show some primary differences between ASMs and LSMs. The simulation results provide the error variations for different methods according to data size and the amount of training time needed for each method. Our

intention is to illustrate the main differences between ASMs and LSMs while providing visualized comparisons for a small-scale power system. There are several reasons why these groups of methods are considered as toy models:

- The majority of the presented works utilized supervised NN to solve the OPF problem, and ANN and CNN are the best candidates to implement.
- Linear regression is a simple method with less complexity and can be analyzed as the first attempt for OPF surrogate models.
- Support vector regression's capacity to handle nonlinearities with a structured risk minimization principle could offer advantages in terms of stability and accuracy over linear regression method. Thus, it is a proper choice for comparison with a linear regression method.

6.1. General Formulations for Methods Involved in Simulations

This section briefly introduces the general formulations for linear regression, support vector regression, neural networks, and convolutional neural networks.

6.1.1. Linear Regression (LR)

LR aims to model the relationship between a dependent variable y and one or more independent variables x . The general form of the model is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon \quad (17)$$

where β_0 is the intercept, $\beta_1, \beta_2, \dots, \beta_n$ are the coefficients, and ϵ is the error term.

6.1.2. Support Vector Regression (SVR)

SVR is a supervised learning model that finds a function $f(x)$ that has at most ϵ -deviation from the actual target values for all training data points. The general formulation is

$$f(x) = \langle w, x \rangle + b \quad (18)$$

subject to $|y_i - f(x_i)| \leq \epsilon$, where w and b define the hyperplane, and ϵ is a tolerance parameter.

6.1.3. Neural Networks

An NN consists of layers of interconnected neurons where the input x is passed through multiple layers to produce an output y . For a simple feedforward neural network, the output is calculated as

$$y = f(W_n f(W_{n-1} \dots f(W_1 x + b_1) + \cdots + b_{n-1}) + b_n) \quad (19)$$

where W_i are the weight matrices, b_i are the biases, and f is an activation function such as ReLU or sigmoid.

6.1.4. Convolutional Neural Networks (CNN)

CNNs are specialized neural networks designed for image and spatial data processing. The core component is the convolution operation, which extracts features from input data. A general convolution layer applies the following operation:

$$h_{ij} = \sum_{m,n} x_{i+m,j+n} \cdot w_{mn} + b \quad (20)$$

where x is the input, w is the convolution filter, and b is the bias term. Multiple convolutional layers followed by pooling and fully connected layers are typically used to produce the output.

6.2. Case Study and Data Generation

The IEEE six-bus system as a standardized test system is widely used in power system studies and considered as the case study to solve OPF via different surrogates (see Figure 4). While larger case studies can be considered, this paper only uses a six-bus IEEE system to showcase the main surrogate models. By using MATPOWER 7.1 the AC OPF is run 10,000 times to generate a labeled dataset. The values of active and reactive demands are changed in the range of $\pm 20\%$ of the nominal values. Only the converged scenarios are included in the dataset, in which the total number of samples in this condition is equal to 9376. The dataset includes the active and reactive demand, active and reactive generated power, voltage amplitude and phase angle for all six buses of the grid. The active and reactive demand of PQ buses are considered as the inputs and the active and reactive generation of the PV buses are selected as the outputs when generating different surrogates. Thus, we have six inputs and six outputs to analyze different surrogate models.

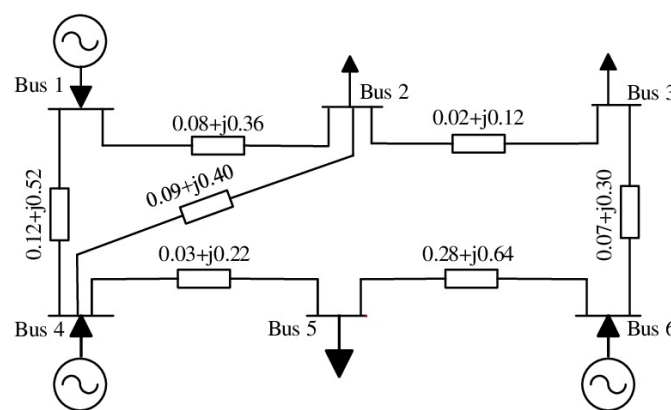


Figure 4. IEEE 6-bus system single line diagram.

6.3. Results

Python 3.10.11 is used to implement the four methods. The simulation system employs an Intel(R) Core(TM) i7-6700HQ CPU @ 2.60 GHz with 8 GB of installed RAM. For LSMs, only one hidden layer with 100 neurons is considered. The activation function is ReLU, the learning rate is 0.001, and the number of epochs is equal to 100. A total of 80% of the data are used as training data and the rest are used for testing. The prediction results for 50 random test samples for all four methods are presented in Figure 5. The ANN method is able to predict the test samples with lower error. Support vector regression is the method with the second-lowest error, followed by linear regression. As is evident, the CNN as an LSM was not able to provide a highly accurate metamodel compared to the others. As shown in Figure 6, the errors of the surrogates are changed based on the size of the dataset. ASMs are less sensitive to dataset size, but LSMs are completely dependent on the number of labeled data. However, the error of LSMs decreases exponentially when more data are fed into the training process. This is one of the reasons that LSMs have gained much more attention recently, since a large amount of data are available. ANN has the best performance with the minimum error in the largest dataset. The ASMs' accuracy is very similar for different dataset sizes. This clarifies the fact that ASMs are more efficient with less available data. On the other hand, the training time for LSMs is linearly increased as the size of the dataset increases, while the training time of linear regression is similar for the entire range of available data. The simulation results show that the ASMs would be more efficient in the event of a shorter amount of training time and reduced availability of data, while LSMs provide better surrogates for AC OPF with less error and high accuracy in the presence of a proper number of labeled data. The LSMs are also able to predict unseen data more accurately, which is very important to guarantee that no constraints will be violated when using the enhanced methods that were reviewed in previous sections.

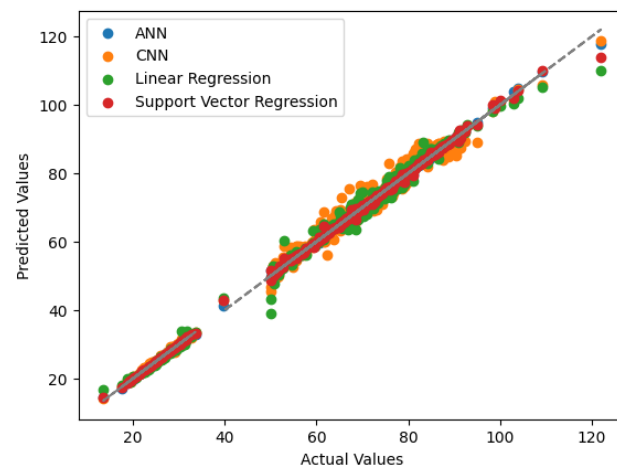


Figure 5. Prediction results for active/reactive power generation for 50 random active/reactive demand samples.

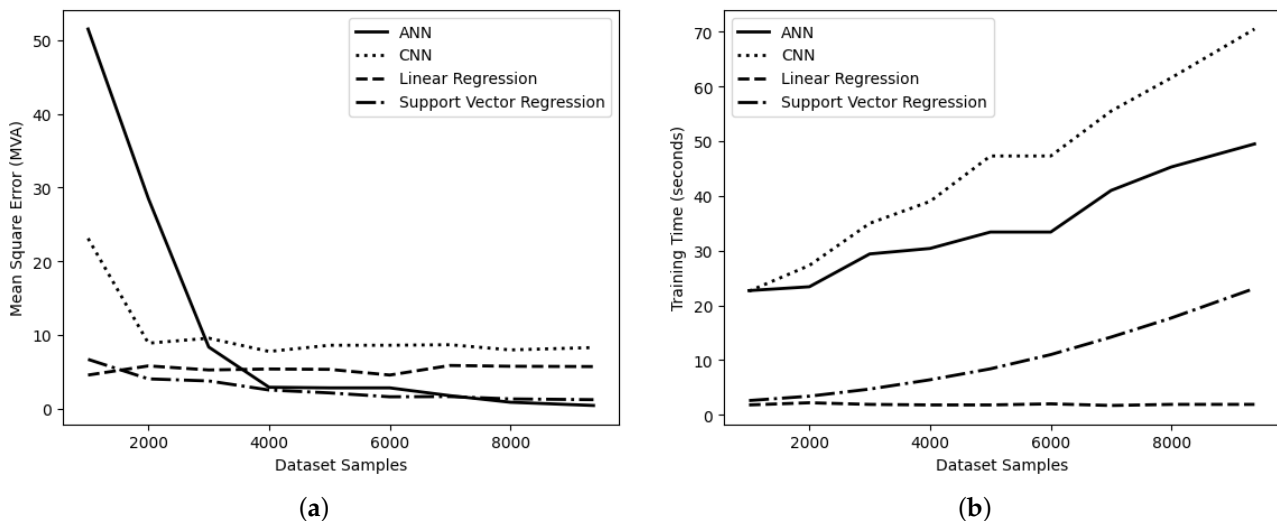


Figure 6. Simulation results for different surrogates: (a) mean square error for different dataset sizes (b) training times for different dataset sizes.

7. Challenges and Future Directions

In this section, the main challenges of these groups are presented. Due to the large input/output size of the end-to-end methods, the training process takes a lot of time and there is an inevitable need to use deep learning-based methods with higher training parameters to extract an accurate meta-model for large-scale power grids. While many physics-informed approaches have been proposed, the generalization of the end-to-end methods remains a major concern. Many of the proposed methods introduced the idea of changing the neural networks based on the physical/engineering constraints of the OPF problem or defining new loss functions to increase the accuracy for unseen data, but changing the neural network structure slows down the training process, especially with regard to external activation commands for a specific neurons and layers. Additionally, defining new loss functions that inherently address the violations requires precise definition and verification for gradient implementation and it may add some extra complexity to the overall surrogate model. Furthermore, generating big labeled datasets is the other main challenge that arises when using supervised end-to-end methods. It means that the OPF problem must be solved for multiple different scenarios before the surrogate model is trained. Also, this approach is not applicable in special conditions where the grid is

not observable, such as instances in which distribution systems or a limited historical data are accessible. The inferential level of the end-to-end methods is still questionable, while many of the proposed methods introduced some efficient solutions. In the other hand, hybrid methods are presented to increase the generalization via minimizing the physical/engineering problem constraints; however, they present some new challenges. Supervised hybrid methods also need big labeled data to facilitate successful prediction. In some of hybrid methods which involve reconstruction, it is necessary to solve nonlinear equations based on analytical methods, which is time consuming for large-scale power systems and does not provide a efficient surrogate. Also, as discussed, multiple hybrid methods utilized multiple neural networks, which increases the complexity of the overall surrogate model and training time. Non-supervised learning methods provide some new opportunities to compensate some of the main drawbacks of the supervised learning approaches, while the generalization and scalability cannot be guaranteed. Overall the main ongoing challenges are as follows:

- Inaccurate random sampling methods covering dependent scenarios for dataset generation and feasible demand samples are not investigated in dataset generation. Non-supervised learning approaches need to consider the feasible/infeasible demand samples, as there is no ground truth that can be applied to check this issue.
- Lack of scalability of the proposed methods.
- Topology changes studies are applicable for only a limited number of variations of the grid topology.
- Low accuracy of NN-based surrogate models in cases of large input/output-size trained models.

Future Directions

As some of the main features have been highlighted in previous subsections, we have revealed multiple avenues for OPF surrogate extraction based on LSMs. While addressing each of the previously mentioned challenges would create new research directions in the field, the authors believe that the ultimate goal of an LSM, as an OPF emulator, would be the extraction of a model that considers all possible scenarios, including large-scale grids with topology changes, contingencies, heavy/light load regime status, and the possibility of predicting all proactive decisions in real-world power systems. However, some future attempts could address the following considerations:

- Reducing input/output size could be enhanced further by integrating modified sampling methods which consider different scenarios for load sample generation. In the majority of presented works, the random sampling method is suggested, which may contain multiple infeasible solutions for OPF problems. As such, the encoder/decoder technique should be applied to decrease the training time and trainable parameters, which is not a efficient approach.
- The GNN would be the best choice for extracting a sufficient metamodel which represents topology changes used to solve the OPF problem. However, a huge labeled dataset is needed to cover all possible topologies that exhibit slight differences from the base topology. One of the solutions would be mitigating the highly correlated geometries by using ML-based classification methods. There is a need to investigate graph attention networks in future studies [103].
- Distribution networks may experience unobserved topology changes because of the lack of efficient meters in the system, undocumented system upgrades and local outages. Grid topology estimation is a way to capture the distribution network which has recently been studied in [104]. This approach is more realistic for OPF surrogates in less observable grids.
- For loss-guided methodologies, it might be useful to consider heuristic methods like GA to handle the back-propagation methods which are non-gradient-based. However, the optimization problem would be very time-consuming for deep networks with a

high number of neurons and layers. Thus, there is a trade-off between discarding the complexity of gradient-based approaches and time-wasting heuristic optimization.

- Unsupervised and self-supervised methods are only implemented in a few works and should be investigated further, since they do not require labeled data. However, the constraint violations would be a significant barrier to using them as proper surrogate models for OPF problems.
- Multi-agent reinforcement learning-based techniques can be used to find the feasible solutions to the OPF problem. While some reinforcement learning-based methods have been introduced to emulate the OPF problem, multi-agent-based scenarios have yet to be evaluated. Additionally, the use of reinforcement learning-based techniques is restricted to warm-start conditions, while future works should explore the applicability of these methods in all considered scenarios. This approach would be a more effective way to study distributed surrogate model extraction in relation to enormous power grids.
- The potential of large language models (LLMs) in optimization problems is underexplored, offering novel capabilities that have yet to be utilized in OPF [105]. Modeling physical constraints is the main challenge for LLM utilization with regard to solving OPF.
- Bayesian neural networks have recently attracted a lot of attention. They inherently consider the probability of predictions. Therefore, they would be a good candidate for classifying critical constraints in OPF problems.
- The interpretability of neural networks is a major concern, and employing explainable AI techniques could enhance the understanding of future LSMs for OPF, like deep symbolic regression methods [106].

8. Conclusions

This paper offers a comprehensive review of the latest surrogate models for the OPF problem, which is instrumental in advancing sustainable power system operations. The surrogate modeling concept is introduced for the first time to categorize the state-of-the-art methods presented for solving OPF. ASMs and LSMs are introduced to employ the main categories for reviewing previous work on this topic. This paper highlights the primary challenges and limitations of ASMs, emphasizing the advantages of LSMs. It further notes the evolving trends and technological advancements that make LSMs more favorable for current and future applications, since the learning part is used to handle unseen data. The LSMs can be strengthened by considering hybrid structure which the hard constraints handled in a better way. This study depicts a clear road map for implementing ASMs and LSMs in the OPF problem with more discussion on LSMs. LSMs are critically analyzed step by step, starting from the simple ideas substituting the Jacobian matrix, considering active sets for constraints and designing new loss functions that are reinforced with constraint violations, reducing the need for labeled data and considering topology changes and contingencies to increase the generalization. Finally, this paper provides a detailed analysis of the main challenges encountered in past research and introduces some key directions for OPF surrogate models at both transmission and distribution levels.

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Abbreviations

OPF	Optimal Power Flow.
DG	Distributed Generation.
ASM	Analytical Surrogate Model.
LSM	Learned Surrogate Model.
GA	Genetic Algorithm.
PSO	Particle Swarm Optimization.
ML	Machine Learning.
CC OPF	Chance Constrained Optimal Power Flow.
NN	Neural Networks.
CNN	Convolutional Neural Network.
KKT	Karush–Kuhn–Tucker
PINN	Physics-informed Neural Networks.
PD	Primal–Dual.
GNN	Graph Neural Network.
GCNN	Graph Convolutional Neural Network.
ANN	Artificial Neural Network.
LR	Linear Regression.
SVR	Support Vector Regression.
LLM	Large Language Model.

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